

VELARIS

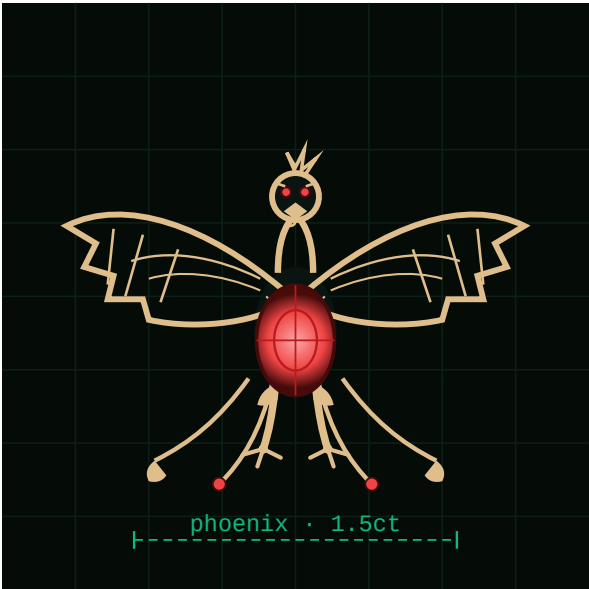
From Inspiration to Jewellery Design

Phoenix Brooch

Generated: September 20, 2026

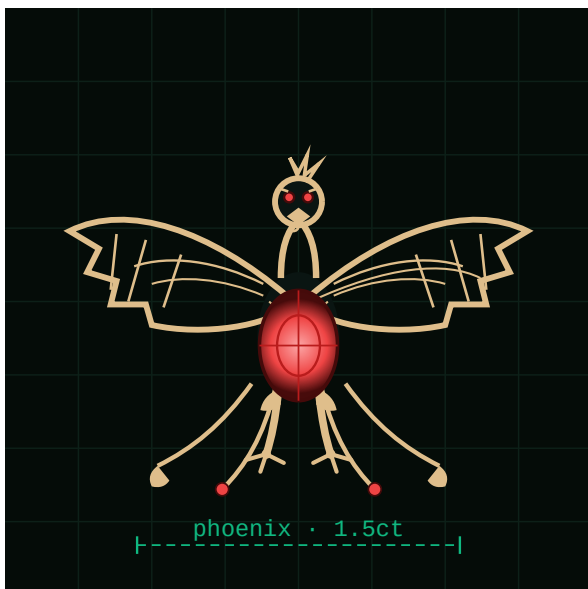
DESIGN VIEWS

Front View



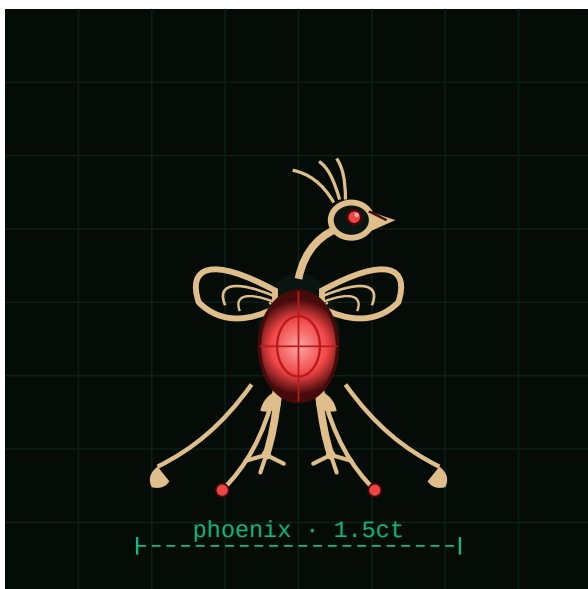
Phoenix front: wings fully spread with layered flight feathers, ruby eyes, upright stance on two taloned hind legs.

Artistic View



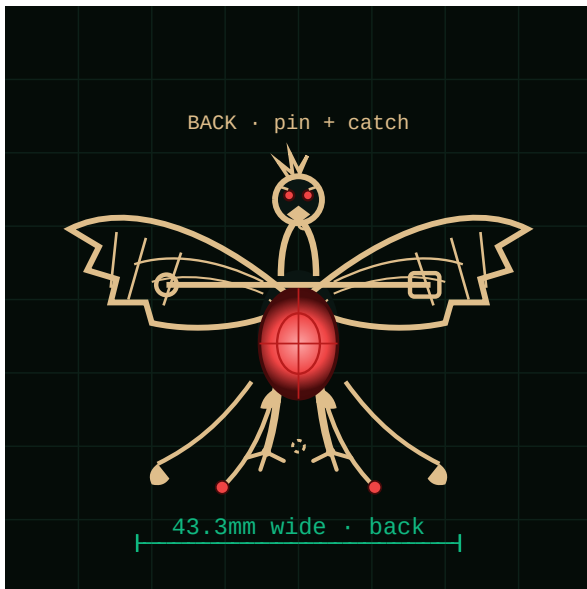
Phoenix at 3/4 angle, near wing raised, depth across breast and tail.

Profile View



Phoenix profile: hooked beak, folded far wing, tail plumes fanning behind the ruby.

Back View



Phoenix back: mirrored silhouette with pin stem, hinge and safety catch.

REALISTIC PREVIEWS

Photorealistic-style studio renders — metal shading, faceted gemstones, soft shadows. For presentation; see CAD DATA for manufacturing geometry.

Front View — Realistic (Procedural preview)



Artistic View — Realistic (Procedural preview)



Profile View — Realistic (Procedural preview)



Back View — Realistic (Procedural preview)



TECHNICAL SPECIFICATIONS

Property	Specification
Design Type	Brooch
Metal	18K Yellow Gold
Gemstone	Ruby
Gemstone Cut	Oval
Gemstone Weight	1.5 ct
Stone Dimensions	6.32 mm × 8.93 mm (W × H)
Overall Width	43.3 mm
Overall Height	49.4 mm
Depth / Thickness	8.0 mm
Band / Frame Width	1.8 mm
Est. Metal Weight	5.6 g
Est. Total Weight	5.9 g
Setting	Prong
Occasion	Anniversary
Stone Size (spec)	1.5 carat
Band Width (spec)	N/A
Details	Phoenix with spread wings, ruby eyes, standing on hind legs

COST ESTIMATION

Cost Component	Amount (USD)
Metal Cost	\$600
Stone Cost	\$2,500
Labor Cost	\$500
Markup (25%)	\$900
TOTAL RETAIL	\$4,500

MANUFACTURING DETAILS

Complexity Score: 68/100

Manufacturing Level: Moderate Complexity



Casting Notes:

18K Yellow Gold casts with exceptional fluidity. For a brooch, the casting also includes a structural pin - cast as part of the main body and reinforced before finishing.

Setting Notes:

Prong setting for Ruby.

Polishing Notes:

Gold two-stage polish with hand-finished recesses.

CAD DATA — MANUFACTURING HANDOFF

Overall: 43.3 mm (W) × 49.4 mm (H) × 8.0 mm (D) · **Stone:** 6.32 × 8.93 mm, 1.5 ct · **Weight:** ~5.9 g total (5.6 g metal)

A DXF file with these true-scale entities (BAND circle, STONE circle, label) is attached to this PDF — open the attachments panel in your PDF reader to send it to Rhino/Matrix/CNC.

DESIGN NARRATIVE

A rampant phoenix brooch with ruby eyes, standing on hind legs.

AI Model: parametric-cad v2 + beauty layer